TOWARDS NET ZERO CARBON BY 2050 - MANET TOWER BLOCK B CASE STUDY, ACCRA GHANA.

Ampomah-Amoako Afua

SUSTAINABLE CONSTRUCTION MANAGEMENT

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1.0 EXECUTIVE SUMMARY

Ghana has an ambition to meet net zero carbon emissions by the year 2050 and aims to reduce greenhouse gas emissions by 8.5 MtCO2e through the implementation of 19 policy actions. The built environment sector is one of the major sectors to be tackled (MESTI, 2021). The aim of this report is to have a retrospective discussion of a case study in Accra towards achieving net zero carbon by 2050.

1.1 Case Study Overview

The chosen project is the Manet Tower Block B located at Airport City in Accra, which serves as the headquarters of the National Investment Bank (NIB).

Name of building: Manet Tower Block B
Type/Use: Commercial/Head office of National Investment bank
Period of construction: 2002-2005
Levels: Eight (8) storey
Total floor area: 5,792 msq. Source: Manet Properties (2011)



Figure 1: Case study project in Accra (Google, 2025)

2.0 SITE APPRAISAL

2.1 Climate Classification: The Koppen climate classification categorises Accra as a tropical savanna climate (Beck et al., 2003).



Figure 2: Location of Ghana in Africa (GISGeophraphy, 2025)

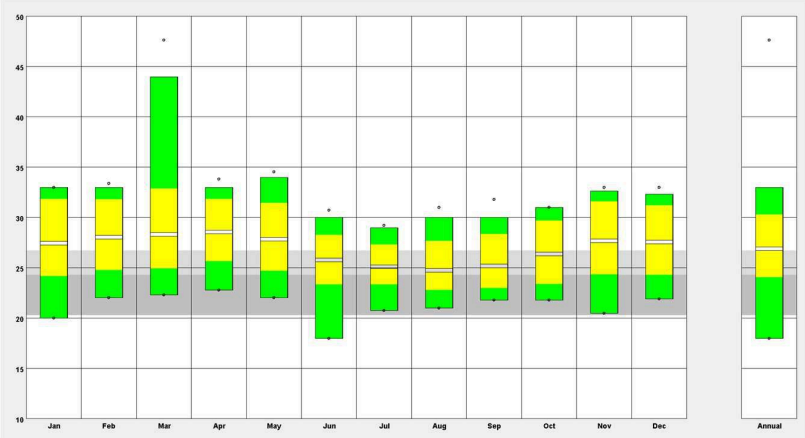


Figure 3: Temperature range of Accra, Ghana. (UCLA Group, 2021)

2.2 SITE OPPORTUNITIES

2.2.1 Microclimate: The site has high solar exposure which is a great opportunity for 3.8 kWH/kWP on-site solar energy according to the World Bank Group (2020). A case study in Singapore, Keppel Infrastructure's building in Changi, has solar PV on the roof and the entire façade producing 600,000 kWh/yr, twice the energy it consumes (Singapore Business Review, 2023).

2.2.2 Building shape and orientation: The case study is currently tilted at an angle, exposing all sides to the eastern and western sun which poses a risk of overheating.

According to Modi et al., (2022), the best orientation in a tropical climate is north, south with shortest sides facing east and west to reduce direct exposure to the sun. According to Sembiring et al., (2024) orientation can account for 6% reduction in total annual energy use, approximately 70kWh which reduces the operational carbon of the building.

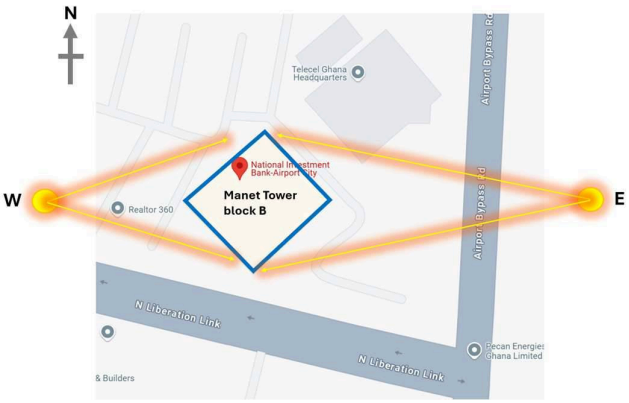


Figure 4: Current orientation of building (Google map, 2025 with modifications by author)

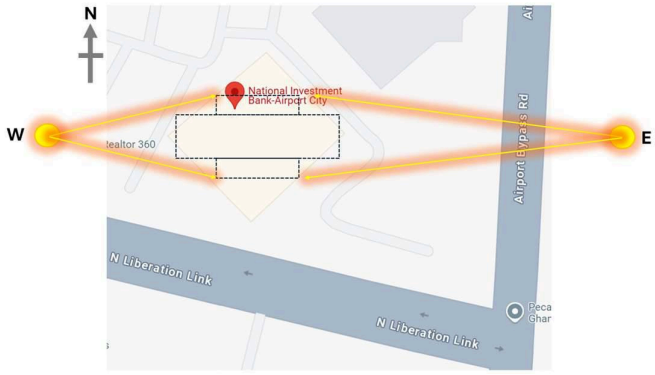


Figure 5: Proposed orientation (Google map, 2025 with modifications by author)

2.2.3 Natural Ventilation and daylighting: The direction of wind at the site is mostly from south and southwest (Climate Consultant, 2024). If the building is oriented towards north and south, the advantage of natural ventilation will be fully realised. A study in the tropical climate of Indonesia proved that natural ventilation, when effectively combined with optimised building envelope can lead to 18.90% reduction in cooling load (Iqbal et al., 2023). This reduces the annual operational carbon of the building.

2.2.4 Façade and material choices: Green roofs and walls reduce the indoor temperature by 17 degrees (Wong et al., 2021). Reflective roofs and walls are effective in reducing indoor temperatures and energy use by 24.3% (Ortiz, 2021). This leads to a lower operational carbon of the building.

Local materials such as laterite bricks, used for the Noomdo orphanage in Burkina Faso, coupled with solar panels for energy generation proves to drastically reduce carbon emissions (Borras, 2024).



Figure 6: Noomdo orphanage, Burkina Faso (Boras, 2024)

2.2.5 Vegetation: Very few trees have been planted at the site but to have better impact, a minimum of 16% of the site should be dedicated to trees to reduce temperature by 1 °C during the day to contribute to cooling effects (Marando et al.,2022).

Dense tree canopies and turf around buildings can provide cooling and lower cooling loads of a building by 2 to 9 degrees (Wong et al., 2021). The Green Walls in Jakob Saigon project in Vietnam successfully incorporated and achieved this.



Figure 7: Green Walls in Jakob Saigon, Vietnam (G8A architects, n.d.)

2.2.6 Subsurface: The subsurface temperature is relatively stable and a great potential for a geothermal system, but thermal imbalance is a challenge regarding their use in tropical countries and may not be fully beneficial (Ratchawang et al., 2022).

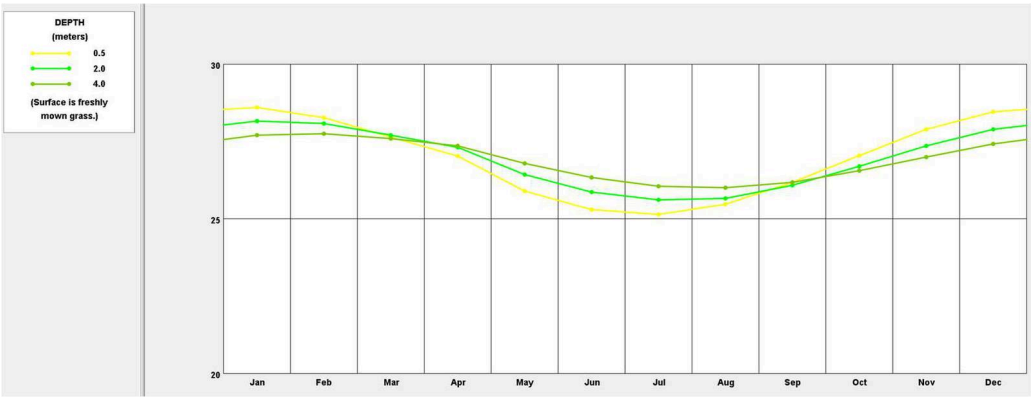


Figure 8: Subsurface temperatures (UCLA Group, 2021)

2.3 SITE CONSTRAINTS

2.3.1 Site boundaries and overshadowing: The case study is overshadowed by adjacent building A. This indicates that the offices within that area will not receive adequate daylighting. According to Essex Planning Authority Association (2022), in preventing overshadowing, a setback of at least one or one and half times the height of the overshadowing building is required. The adjacent building's overshadowing will negatively impact future rooftop solar PV and building integrated PV energy generation.

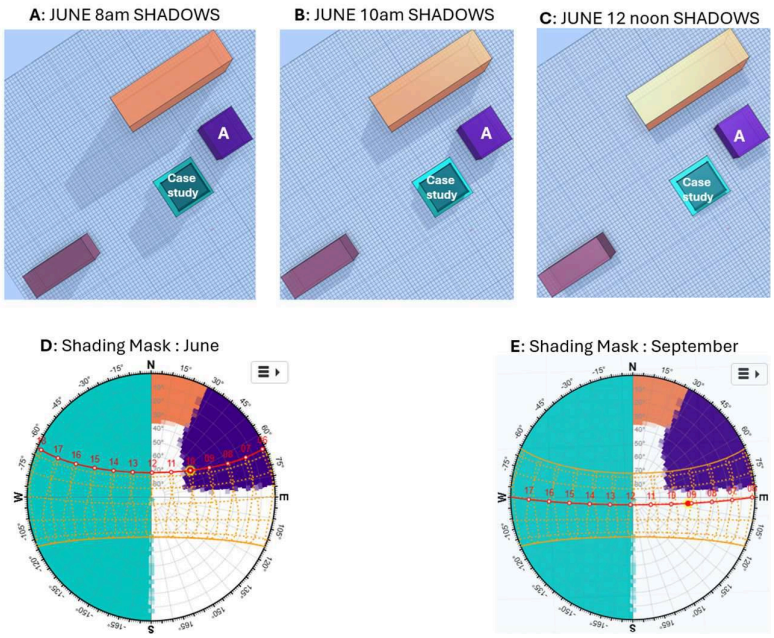


Figure 9: Demonstration of overshadowing on site (Marsh, 2025)

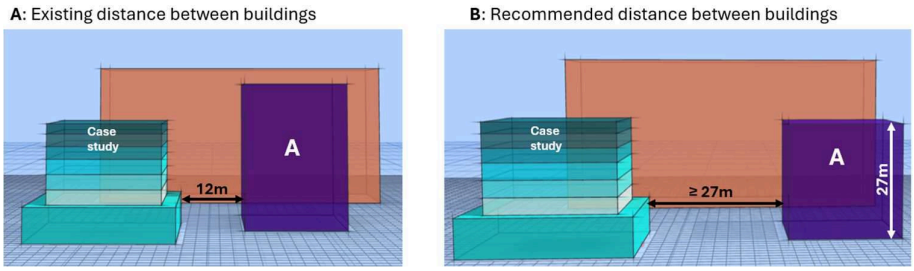


Figure 10: Current and proposed setback of case study (Marsh, 2025)

2.4 Power: an opportunity and limitation

The building is totally reliant on the national grid which is partly thermal and hydro but very minimal solar power. The grid is readily available for the project to connect but frequent power outages makes this a big challenge and projects sometimes rely on diesel powered generators.

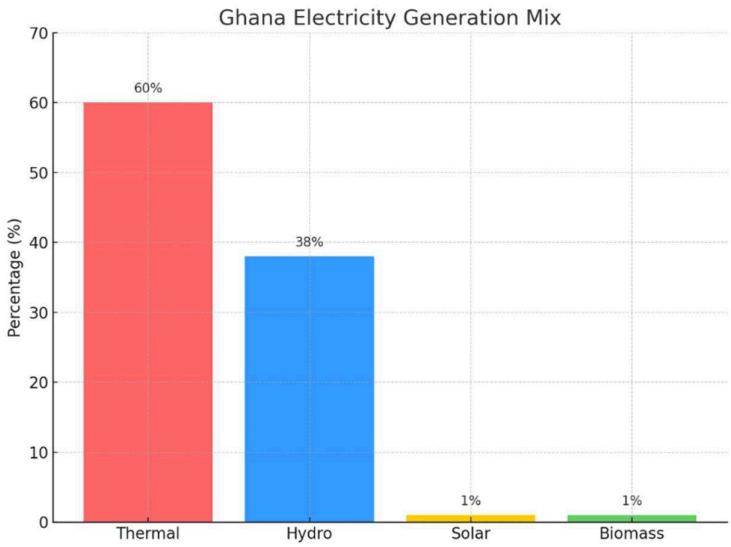


Figure 11: Ghana's energy generation (Acquah, 2024)

Solar is a possibility on the rooftop as used in the Ken TTDI building in Kuala Lumpur. Ghana has plans to establish a nuclear power plant as a source of renewable energy by 2034 to be able to achieve net-zero by 2050 (Ghana Broadcasting Corporation, 2025). This will lead to the possibility of purchasing Renewable Energy Guarantees of Origin (REGO) to offset the building's huge cooling needs.

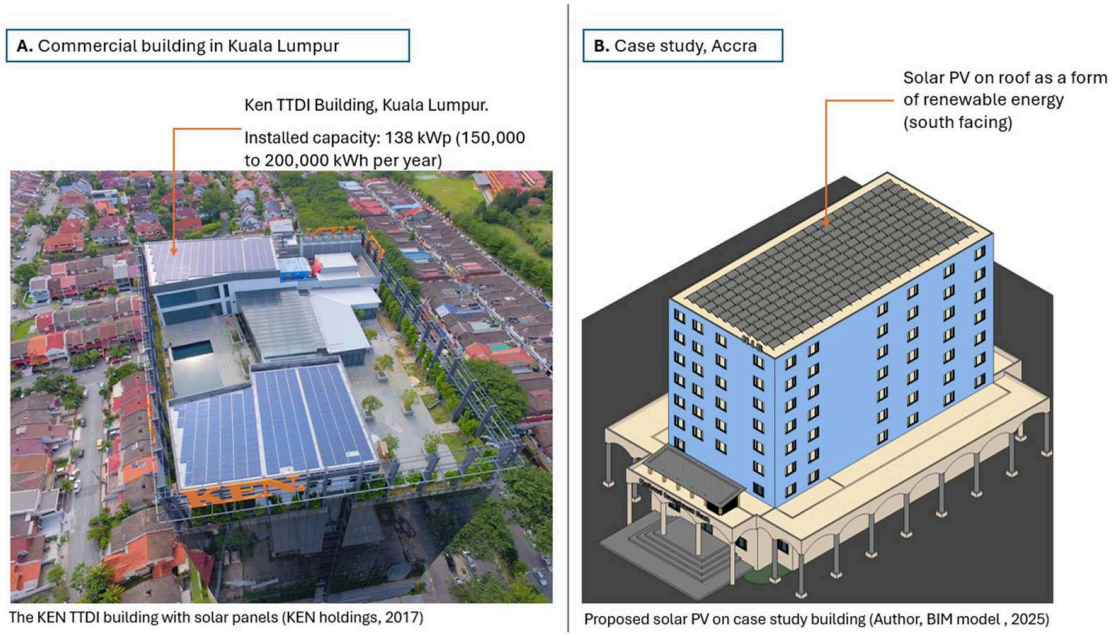


Figure 12: Ken TTDI building in Kuala Lumpur with solar panels and proposed for case study building

2.5 Energy Use

The building has an estimated energy use intensity of 180 kWh/m^2/yr, based on similar projects in Singapore and Malaysia (Al-Masrani et al., 2018). 56% of energy in tropics is dedicated to cooling systems (Katili et al, 2015).

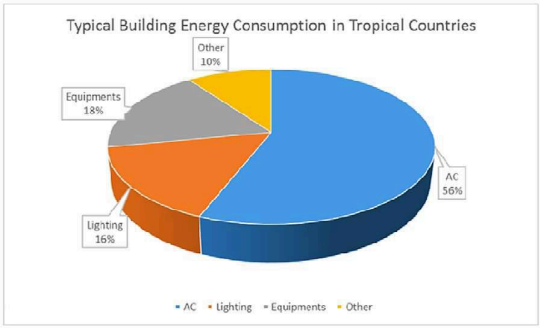


Figure 13: Breakdown of energy use (Katili et al, 2015)

Summary: Vegetation, orientation and natural ventilation have some degree of impact on reducing operational carbon of a building. Apart from the these, passive design strategies could effectively reduce the cooling demand and lead to reduced operational carbons (Adegbie, 2021). Facades which have low U-values and energy efficient cooling systems can lead to 36% reduced energy consumption and maintain occupant comfort (Kalwry and Atakara(2025).

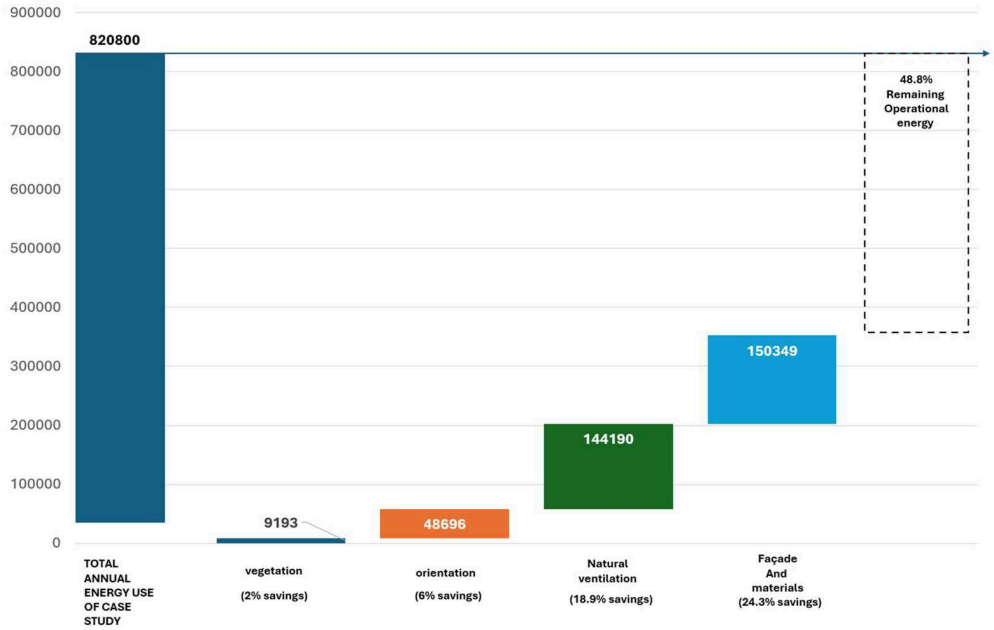


Figure 14: Impact of site strategies on achieving net zero (Author, 2025)

3.0 NET ZERO CARBON DESIGN PRINCIPLES

3.1 Passive Design

Effectively combining passive strategies such as orientation, natural lighting and ventilation can lead to a reduction in energy consumption by 40% (Falzon et al., 2023).

3.1.1 Building Orientation

In achieving net zero in buildings, passive design strategies that are critical to consider in tropical climates include orientation, natural ventilation and wall-to-window ratio to reduce energy use by 34%-35% (Ohene *et al*. (2024).

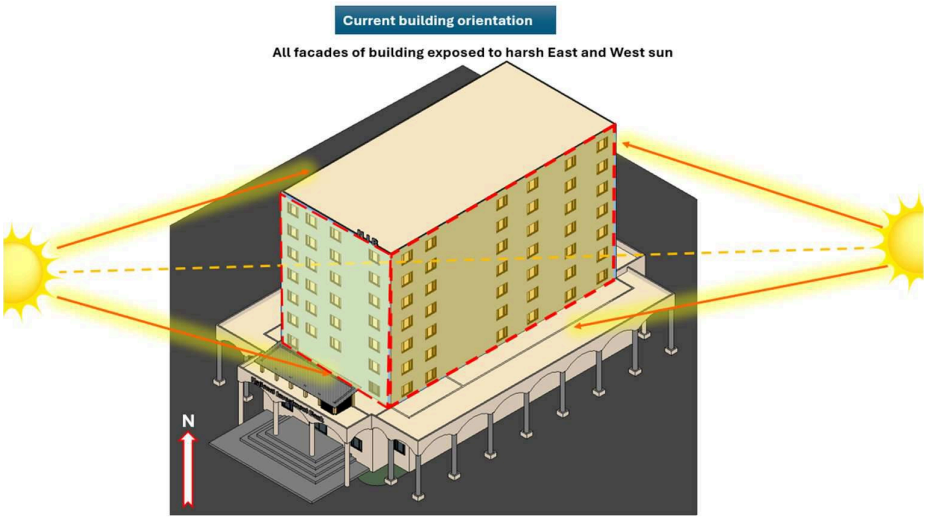


Figure 15: Impact of the sun on current orientation (Author, 2025)

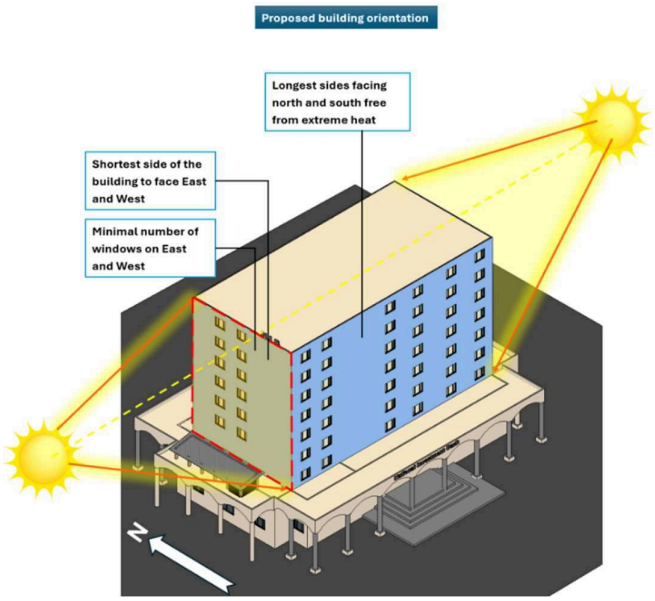


Figure 16: The best orientation - minimal impact from the sun (Author, 2025)

3.1.2 External shading

Horizontal and vertical shading devices are effective in reducing solar heat gain (Lenoir, 2013). Dynamic solar photovoltaic shading devices with different moving strategies such as sliding or rotation can contribute to energy savings of 50.38% (Yan *et al.*, 2024).

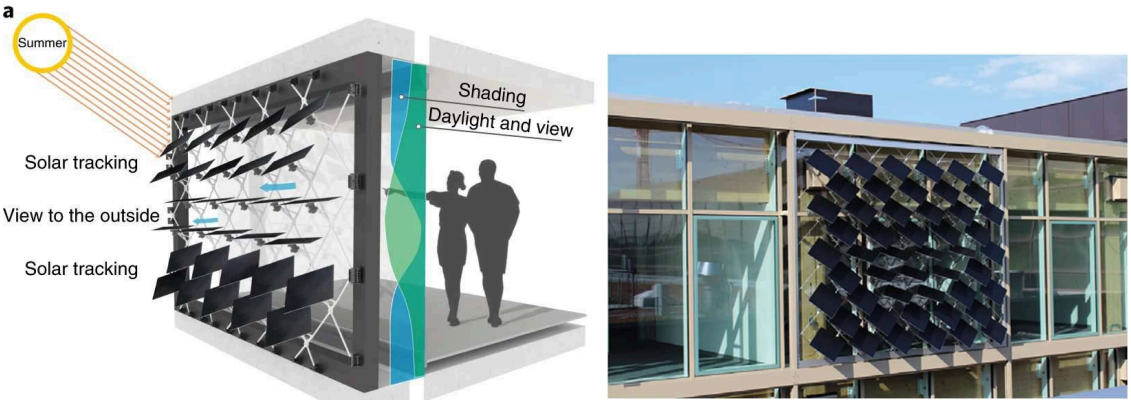


Figure 17: Dynamic solar PV shading at the House of Natural Resources in Zurich (Schlüter, 2015)

3.1.3 Natural ventilation and cooling

Cross, stack ventilation, courtyards and atriums are effective passive cooling methods to reduce cooling loads (Lenoir, 2013). One airport square in Accra and EastGate building in Zimbabwe incorporate courtyards and atriums that aid stack effect to remove heat from the building.

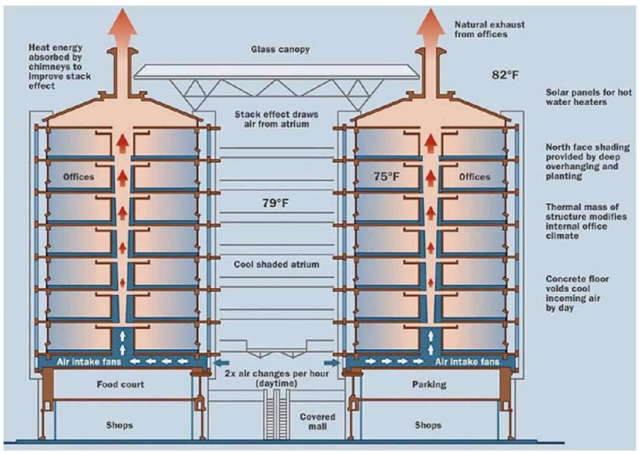


Figure 18 : Shading and cooling strategies - EastGate centre project, Harare (Pearce, n.d.)

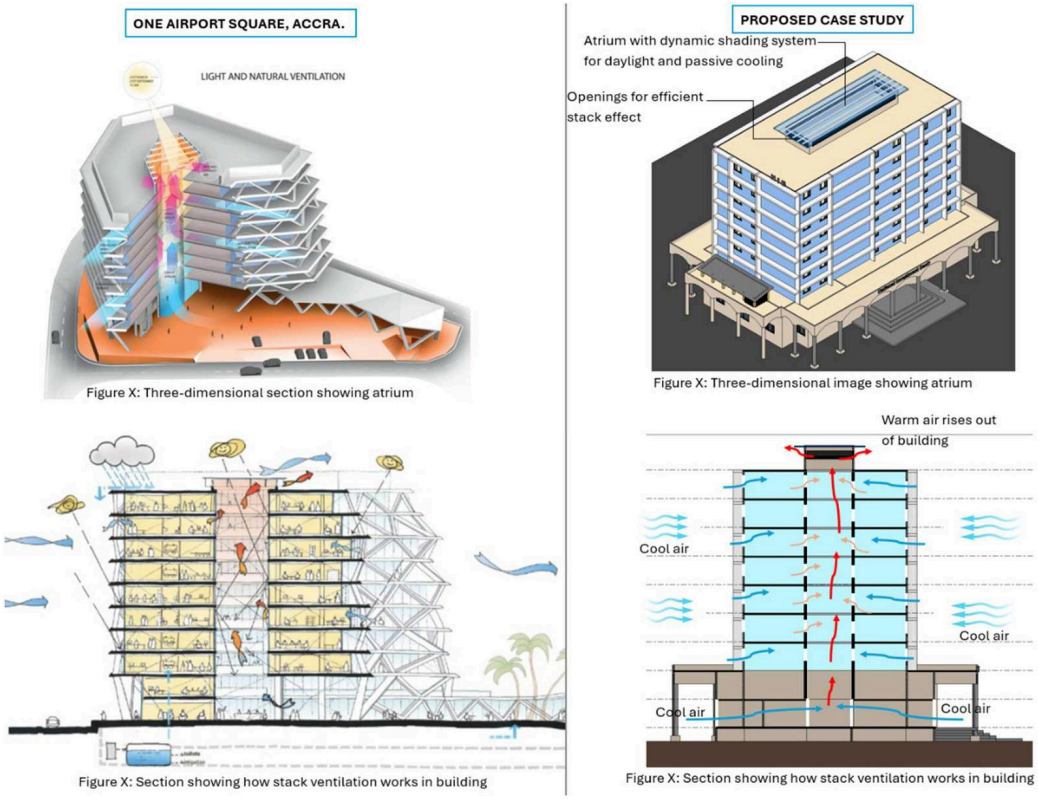


Figure 19: One airport square project in Accra and how an atrium could have been incorporated in the case study

3.1.4 Daylighting

Atriums and light shelves coupled with reflective ceilings create uniformly distributed daylighting in office buildings and save energy during the daytime by 46% (Binarti and Dewi, 2016).

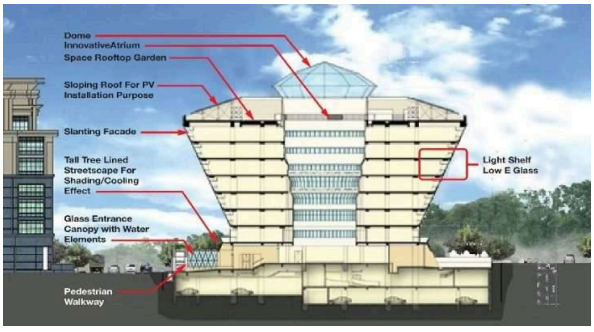


Fig. 20: Atrium for diffused light in Malaysia's St Diamond Building (Energy Commission, Malaysia, n.d.)

3.1.5 Building Fabric

Locally sourced materials should have been considered for the project (Table 1). Low-E glazing windows, cool roofs and exterior colours with high albedo are effective in minimising heat gain.

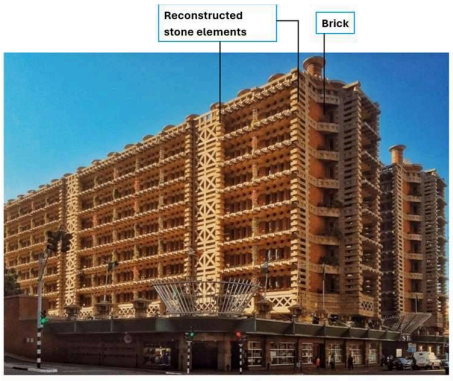


Figure 21: EastGate centre fabric, Harare, Zimbabwe (Pearce, n.d.)



Figure 22: Proposed view of case study (Author, 2025)

3.2 Materials

Table 1: current materials for case study versus low carbon options

Materials used	Low carbon options common in Ghana
Concrete and steel	Timber
Glass curtain walls	Stabilized laterite
Aluminium cladding	Clay brick walls
	Recycled steel

(Author, 2025 with information from Circular Ecology, 2019)

Timber and laterite are sustainable materials that are common in Ghana. The compressive and tensile strength of timber makes it an excellent material for building (Dotel, 2024). Using timber is efficient and sustainable as demonstrated by the Mjøsa Tower in Norway (Tupėnaitė et al.,2020). The Monadnock building is a17-storey building with brick façade and steel structure to firmly hold the building in place (Oldfield, 2019).

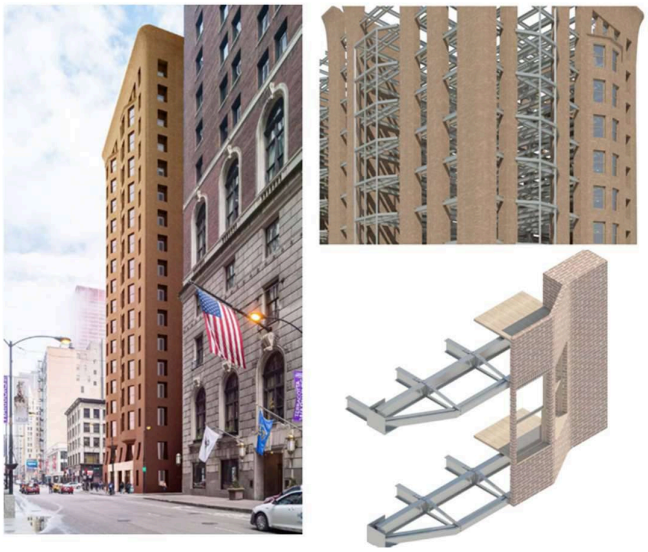


Figure 23: Monadnock building in Chicago (Matteis, 2018 p.69-71)



Figure 24: Premier office in Vietnam (Abdel, 2023)



Figure 25: Mjosa tower with timber structure (Pintos, 2020)

3.3 Summary: Passive design strategies can reduce the huge cooling needs in tropical regions leading to reduced operational energy. As previous studies have shown, passive design strategies reduce carbon emissions by 34-50.38%, it wouldn't make the building completely net zero but contributes a fraction towards reducing carbons.

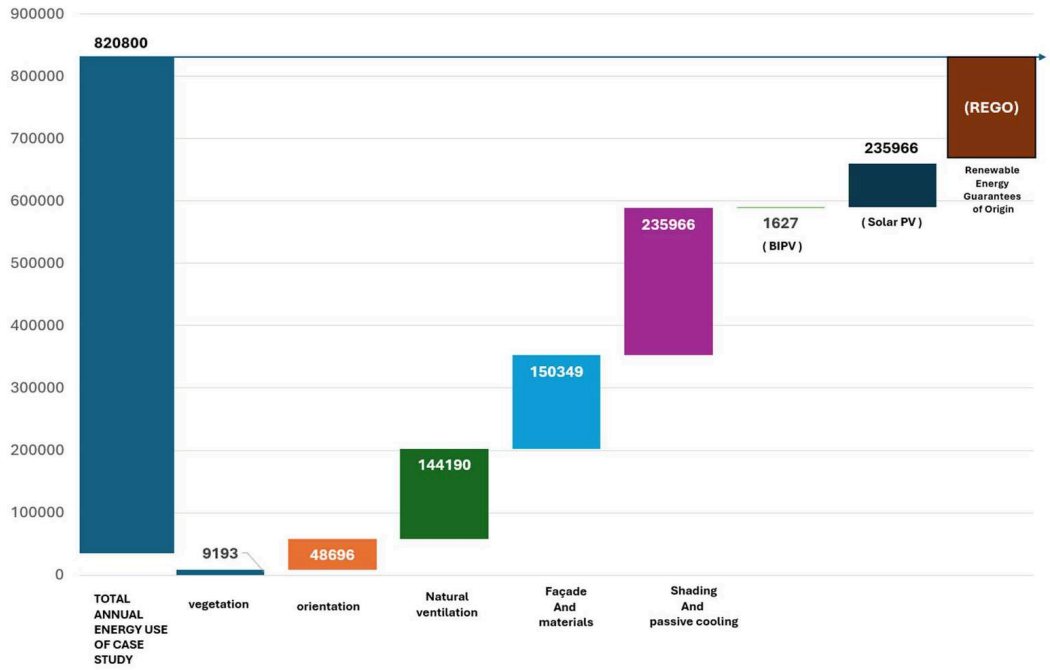


Figure 26: Summary of impact of site, passive strategies and renewable energy in achieving net zero (Author, 2025)

4.0 WHOLE LIFE CARBON ANALYSIS

Whole life net zero carbon is defined by the UKGBC (2019) as when the total carbon emissions associated with the operational and embodied carbon of a building has an impact over its' lifetime is zero or negative. In achieving this, embodied, operational and the afterlife carbon choices of the building must be considered.

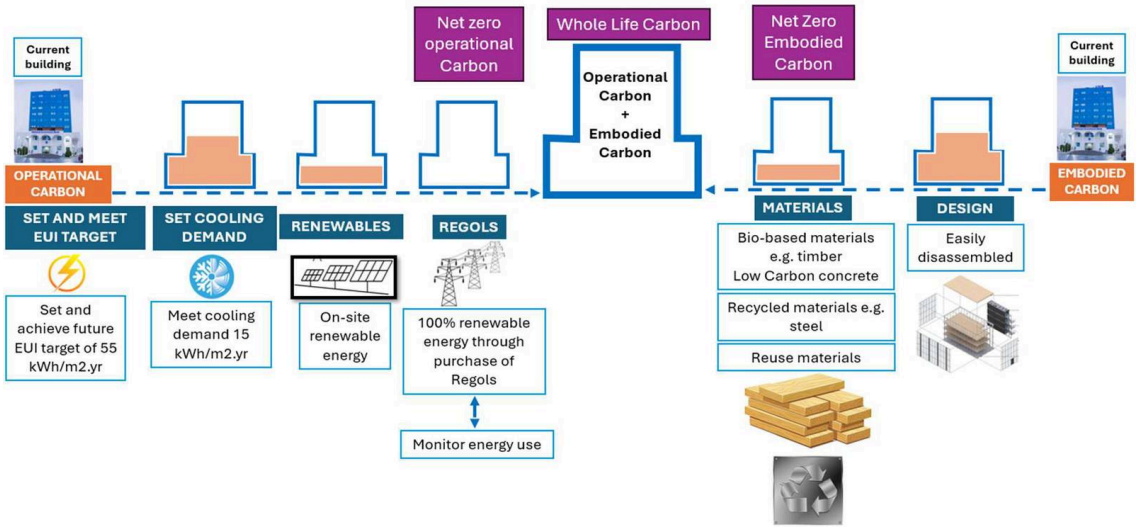


Figure 27: Whole life carbon explained (LETI Guide, 2020)

Retrospectively, the whole life carbon of the case study could be calculated at the onset of the project using the underlisted tools.



Figure 28: tools to calculate whole life carbon emissions (Finnegan, 2025)

4.1 Fabric First Approach

The first step in dealing with net zero emissions is fabric first approach together with passive design to reduce solar heat gain (UKGBC, 2019). Fabric should have low U-values (LETI Guide, 2020).

Compressed earth blocks is an excellent low carbon choice as it is a high-performance envelope. Shaded low-emissivity glass window with low solar heat gain coefficient should have been the best window option as demonstrated in a study of indoor thermal environment in office buildings in Nigeria by Adegbie (2021) which informs an 89.54% improvement in indoor thermal environment.

4.1.1 Materials

Table 2: Carbon footprints of current and proposed materials

Building element	**Current materials**	**Carbon footprint**	**Proposed materials**	**Carbon footprint**
Structure	Reinforced concrete	350–500 kg CO_2e/m^3	Timber frame	100–150 kg CO_2e/m^3
Roof	Concrete slab with bitumen membrane	200–350 kg CO_2e/m^3	High albedo roof with Limestone Calcined Clay Cement slab	210 kg CO_2e/m^3
External cladding	Curtain wall with aluminium frames	120–185 kg CO_2e/m^2	Compressed earth blocks	5–40 kg CO_2e/m^2
Wall infill	i)Sandcrete blocks ii) gypsum board	i)140–300 kg CO_2e/m^3 ii)	i)Compressed earth blocks ii)Wood fibreboard	i)5–40 kg CO_2e/m^2

(Fernandes et al., 2019) (Arup, 2023) (Circular Ecology, 2019)

4.2 Operational carbon emissions

According to the UKGBC (2019), operational carbon refers to carbon emissions that is related to heating, ventilation, cooling, lighting and other electrical equipment. This can be achieved by reducing the energy demand and substituting any form of fossil fuel energy with 100% renewable energy through purchase of Renewable Energy Guarantees of Origin (REGOs).

Step 1: Set an Energy Use Intensity (EUI) Target for project

The case study should target an EUI of 55 kWh/m2.yr (LETI Guide, 2020) and reduce energy for cooling in a tropical setting to 15 kWh/m2.yr.

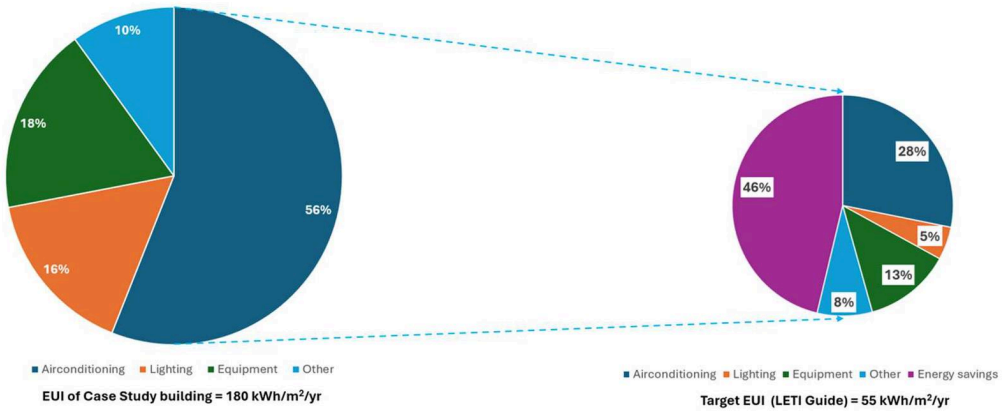


Figure 29: Space cooling targets (LETI Guide, 2020 p.46)

Step 2: Efficiency of building systems

After setting the energy use target, considering that cooling demand in the tropics is high, having efficient HVAC systems such as the Variable Refrigerant Flow (VRF) systems which can lead to 26.43% improvement in energy efficiency, eventually leading to reduced CO2 emissions (Hussien et al., 2023).

4.2.1 Renewable energy – onsite solar photovoltaics

Solar PV is the best form of renewable energy because Ghana has high solar radiation. Even though this is an onsite renewable option, it will be very difficult to harvest enough energy that will totally keep balance the operational energy of the building.

4.3 Embodied carbon

The following steps are important to achieve lower embodied carbons:

Step 1: Have embodied carbon emissions targets

The LETI Guide (2020) suggests setting a target of less than 350kgCO2/m2 for non-domestic buildings. Huge carbon reductions can be achieved if the superstructure and material selection consider low carbon choices.

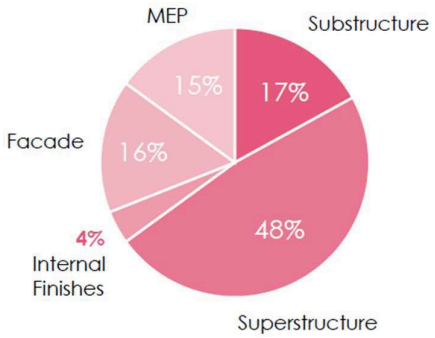


Figure 30: Embodied carbon breakdown (LETI Guide, 2020 p.62)

Step 2: Reduce embodied carbon with low carbon materials

i. **Low carbon substructure and super structure**

Selecting low carbon materials for this part of the project is important. For the substructure which would most likely incorporate reinforced concrete, ordinary Portland cement can be replaced with low carbon concrete. Timber structural frame is the best material as this is abundant in Ghana and has been used in similar projects like the Mjøsa Tower in Norway.

ii. **Low carbon building roof and façade materials**

Comparing carbon emissions of different materials is crucial and can be sought from Environmental Product Declarations (EPDs). Suggested facade materials are using low compressed earth blocks and terracotta panels.

iii. **Mechanical Electrical and Plumbing (MEP)**

Incorporating efficient MEP systems such as VRF systems, smart building management systems that controls the use of HVAC and efficient lighting such as LEDs.

iv. **Finishes, Furniture, Fixtures and Equipment:**

The project should incorporate timber for finishes as it has very low carbon emissions. All products should have their EPDs carefully analysed to avoid high carbon emission materials and harmful chemicals.

v. **Design for Manufacture and Assembly (DfMA)**

Considering the building as a product that can be easily disassembled and reused is the best option for the future. The case study should have considered modular designs and prefabricated building elements with low carbon materials. In achieving the set target, the project should consider low carbon material selection and 50% of materials from recycled sources. At the end of the life of the project, all materials should have a re-use rate of 80%.

4.4 Summary: It is challenging to achieve 100% embodied carbon due to the percentage of building materials (about 60%) that are imported from various parts of the world to Ghana. Incorporating technology should be further considered in achieving net zero.

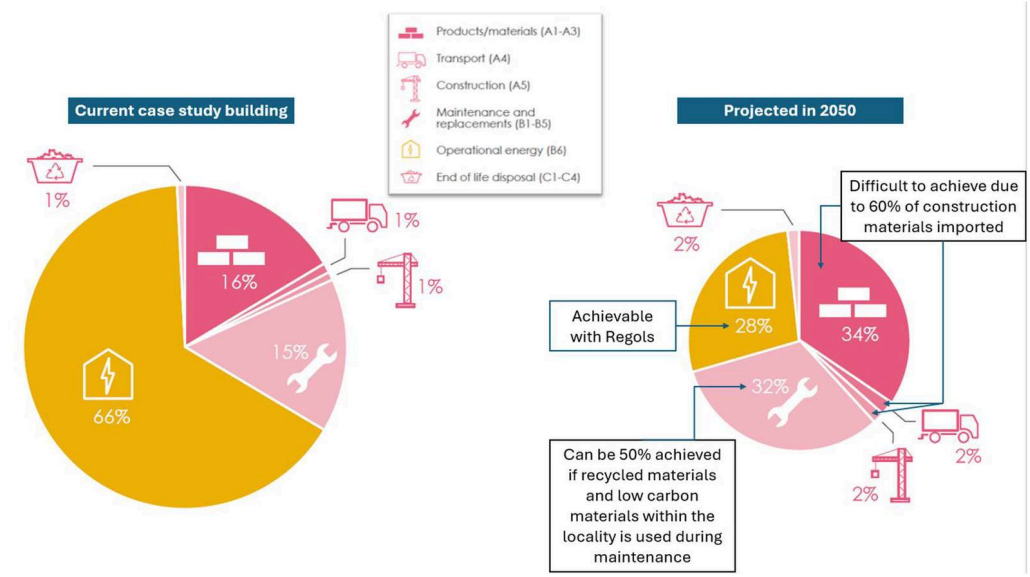


Figure 31: Concluding whole life carbon of project (LETI Guide, 2020 p. 59)

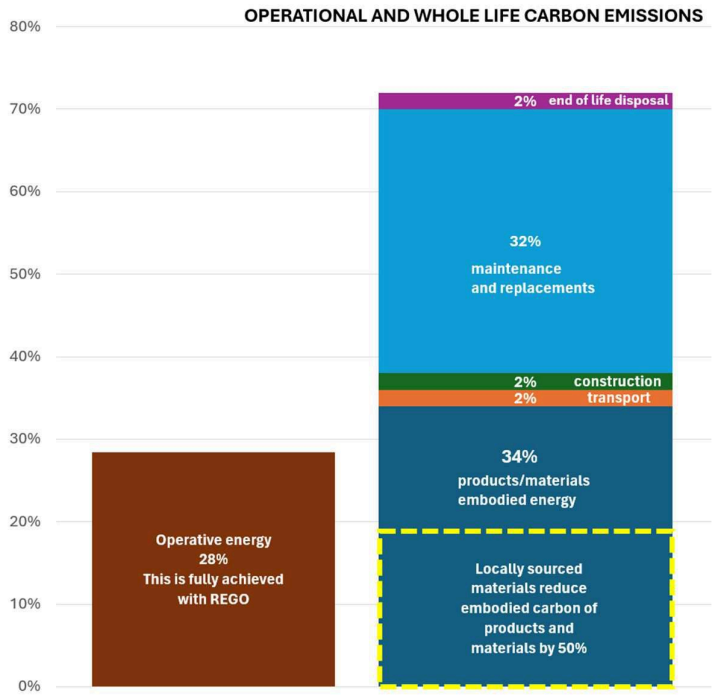


Figure 32: Operational versus whole life carbon emissions (Author, 2025)

5.0 TECHNOLOGY

Sustainable technologies according to Finnegan (2018) refers to how a building is ventilated, powered and heated or cooled using specified equipment that are known not to have prolonged adverse carbon dioxide impact on the environment.

Table 3: Renewable sustainable technologies for project

RENEWABLE SUSTAINABLE TECHNOLOGIES	Suitability for case study	Reason
Solar PV	Yes	Ghana has a high solar radiation for solar energy
Wind	No	Average wind speed for urban Accra is low
Hydro	No	Location and cost of hydro is for large scale cities but not individual buildings
Energy from waste	No	Waste-to-energy infrastructure is non-existent in Ghana
Kinetic flooring	No	Good technology to be placed at the entrance of the bank but not recommended because of high initial cost but low net energy return
LED Lighting	Yes	Contributes to low energy use
Building Management Systems	Yes	Complements HVAC and lighting systems to reduce energy demand.

Source (Author, 2025)

5.1 Solar PV: Solar PV to be located on the building's 460 m2 flat roof. The roof also houses outdoor units and leaves a net roof space of 230 m2 for the solar panels.

Table 4: Total solar PV energy and cost calculation

Roof Area	Projected solar PV calculation	Case study energy consumption	Remarks
230 m2	Assuming using 550W commercial panels :104 panels x 550 W = 57.2kW	Annual energy use of the Manet building is 250,800 kWh/year based on LETI EUI for office buildings	Solar PV will produce 30% of the total annual energy use of building.
	Annual energy the system will generate is 75,100 kWh/year		
	Cost of installation of this system with batteries is approximately $134,000	Cost of electricity from the grid is £0.14 to £0.18 per kilowatt-hour Cost per year = $59,070.92	

(Source: Author, 2025)

The solar panels if installed will reduce 30% of the operational energy of the building whiles increasing the embodied carbon of the building. Panels are mostly imported from China to Ghana and this will further increase the whole life carbon of the building. This will be a good addition to the building later in future when Ghana's grid allows for net metering for the building to monetize solar energy.

5.2 Light Emitting Diode (LED) lighting technology: LED lighting is energy efficient and contributes to 35% cost savings per month than fluorescent lamps when successfully combined with Building Management Systems (Murdiyansyah et al.,2024). There is a reduction of carbon emissions when LED lighting is used instead of fluorescent lights (Stravoravdis, 2013).

5.3 Conclusion: Technology has a role towards achieving net zero in buildings but does not lead to 100% net zero carbon in buildings. For the suggested technologies of solar PV and LED lighting discussed above, the amount of energy solar PV will produce for the building will not be able to make up for the entire operational energy of the building considering the total cost of installation.

It will be possible to achieve 100% operational carbon emissions through REGO but the percentage of embodied carbon emissions related to materials, construction and transportation will be 50% achieved. In this case, we wouldn't have achieved 100% net zero whole life carbon leading us to the last option of carbon offsetting.

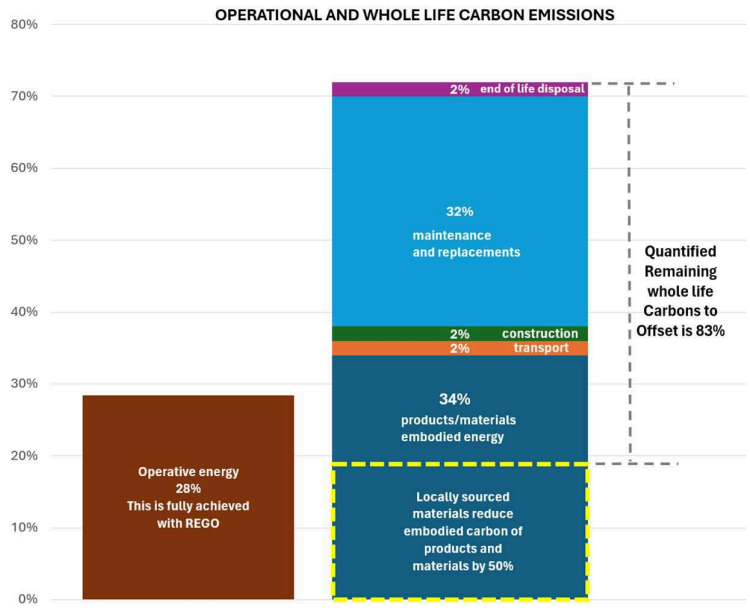


Figure 33: Quantified whole life carbon for offsetting (Author, 2025)

6.0 Carbon offsetting

Carbon offsetting requires investing in projects that reduce or remove carbon, and this could be in emissions reductions or carbon removals. According to the University of Oxford offsetting principles, even though reducing emissions is a basic step, gradually increasing investments towards carbon removals is the best. The Keppel bay tower project after involving all strategies and energy efficient systems, purchased carbon offsets for the remaining energy since it could not achieve 100% net zero energy (Reuters, 2024) Keppel bay tower project link.

Considering the available options, residual emissions will be compensated for with the purchase of carbon credits that invests in direct carbon removal to the geosphere which has several benefits and low reversal risk, examples are Biochar and Direct Air Carbon Capture and Storage (DACCS).

Step 1: **Quantify residual carbon emissions**: this can be calculated with One click LCA.

Step 2: **Budget for carbon removal**: A projected amount should be specified, and a projection can be obtained from similar projects.

Step 3: **Purchase the carbon removal option**: The best option to invest in is the Direct Air Capture (DAC) with providers like Climaworks.

Step 4: Finally, the exact amount of purchased offset in Step 3 should be made public in line with reporting procedures to prove this with data to avoid greenwashing.

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